

EXPERIMENT-12

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Date of Performance: 21/9/26

Q1. DESIGN A TRIGGER WHICH:

**--1. WHENEVER THERE IS A INSERTION ON STUDENT TABLE THEN, THE CURRENTLY
INSERTED OR DELETED**

--ROW SHOULD BE PRINTED AS IT AS ON THE OUTPUT CONSOLE WINDOW.

```
CREATE OR REPLACE FUNCTION fn_student_audit()
RETURNS TRIGGER
LANGUAGE plpgsql
AS
$$
BEGIN
    IF TG_OP = 'INSERT' THEN
        RAISE NOTICE 'Inserted Row -> ID: %, Name: %, Age: %, Class: %',
            NEW.id, NEW.name, NEW.age, NEW.class;
        RETURN NEW;

    ELSIF
        TG_OP = 'DELETE' THEN
        RAISE NOTICE 'Deleted Row -> ID: %, Name: %, Age: %, Class: %',
            OLD.id, OLD.name, OLD.age, OLD.class;
        RETURN OLD;
    END IF;

    RETURN NULL;
END;
$$
```

Q2. DESIGN A PORSTGRESQL TRIGGERS THAT:

Whenever a new employee is inserted in `tbl_employee`, a record should be added to `tbl_employee_audit` like:

"Employee name <emp_name> has been added at <current_time>"

Whenever an employee is deleted from `tbl_employee`, a record should be added to `tbl_employee_audit` like:

"Employee name <emp_name> has been deleted at <current_time>"

The solution must use PostgreSQL triggers.

```
CREATE OR REPLACE FUNCTION audit_employee_changes()
RETURNS TRIGGER
LANGUAGE plpgsql
AS
$$
BEGIN
    IF TG_OP = 'INSERT' THEN
        INSERT INTO tbl_employee_audit(message)
        VALUES ('Employee name ' || NEW.emp_name || ' has been added at ' ||
NOW());
        RETURN NEW;

    ELSIF TG_OP = 'DELETE' THEN
        INSERT INTO tbl_employee_audit(message)
        VALUES ('Employee name ' || OLD.emp_name || ' has been deleted at ' ||
NOW());
        RETURN OLD;
    END IF;

    RETURN NULL;
END;
$
```

```
CREATE TRIGGER trg_employee_audit
AFTER INSERT OR DELETE
ON
tbl_employee
FOR EACH ROW
```

```
EXECUTE FUNCTION audit_employee_changes();
```

```
--TESTING THE TRIGGER
```

```
-- Insert an employee
```

```
INSERT INTO tbl_employee(emp_name, emp_salary) VALUES ('Aman', 50000);
```

```
-- Delete an employee
```

```
DELETE FROM tbl_employee WHERE emp_name = 'Aman';
```

```
-- Check audit log
```

```
SELECT * FROM tbl_employee_audit;
```